Estimators and Estimation

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Introduction

An **estimator** is a rule – a function of the data – that produces a guess about an unknown parameter of the population. An **estimate** is the value that rule returns on a specific sample. The sample mean is an estimator of the population mean; on the sample $\{3.1, 2.7, 3.4\}$ it returns the estimate 3.07. Every inferential procedure begins with an estimator.

Prerequisites

The reader should know what a sample is, what a parameter is, and should be comfortable calling R functions on a numeric vector.

Theory

Given iid observations $X_1, \ldots, X_n$ from a distribution F with parameter θ, an estimator is any function $\hat{\theta} = T(X_1, \ldots, X_n)$. Common examples:

- Sample mean $\bar{X} = \frac{1}{n} \sum X_i$ for the population mean μ.
- Sample variance $s^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ for σ^2.
- Sample proportion $\hat{p} = \bar{X}$ for π when $X_i \in \{0, 1\}$.
- Sample correlation $\hat{\rho}$ for ρ.

The **plug-in principle** motivates many estimators: to estimate any functional $\theta = T(F)$ of the distribution, replace F with the empirical distribution $\hat{F}_n$ and compute $T(\hat{F}_n)$. This principle yields the sample mean as an estimator of the population mean, the sample quantile as an estimator of the population quantile, and so on.

Properties by which estimators are judged:

- **Unbiasedness**: $E[\hat{\theta}] = \theta$.
- **Consistency**: $\hat{\theta} \to \theta$ in probability as $n \to \infty$.
- **Efficiency**: small variance (ideally attaining the Cramer-Rao lower bound).
- **Robustness**: stability under deviations from modelling assumptions.
- **Sufficiency**: using all the information about θ in the data.

General-purpose construction methods – method of moments, maximum likelihood, least squares, Bayesian posterior summaries – each yield estimators with characteristic properties under the model they assume.

Assumptions

The plug-in principle is model-free in spirit but model-dependent in application: the estimator inherits any bias from the mismatch between the assumed model and reality. When the sample does not represent the target population, no estimator can rescue the inference.

R Implementation

```r
set.seed(2026)
x <- rnorm(40, mean = 75, sd = 12)

mean(x)
var(x); sd(x)
median(x); quantile(x, probs = c(0.25, 0.5, 0.75))
mean(x > 70)
cor(x, x + rnorm(length(x), sd = 5))
```

Each call is a plug-in estimator: `mean()` for μ, `var()` for σ^2, `quantile()` for population quantiles, `mean(x > 70)` for $P(X > 70)$, `cor()` for ρ.

Output & Results

```
[1] 73.80       # sample mean
[1] 152.37      # sample variance
[1] 12.34       # sample SD
[1] 73.87       # sample median
     25%     50%     75%
   65.13   73.87   81.62
[1] 0.60       # plug-in estimate of P(X > 70)
[1] 0.922      # sample correlation
```

Each of these is a point estimate of the corresponding population quantity; none tells us how uncertain that estimate is.

Interpretation

A point estimate is not enough. Every estimate reported in a paper should be accompanied by a measure of its uncertainty – a standard error, a confidence interval, or a credible interval. "Mean systolic blood pressure 128 mmHg" is a statement; "128 mmHg (95% CI 125–131)" is inference.

Practical Tips

- The plug-in principle gives a good default estimator for most quantities; check for bias only when you have reason to suspect it.
- For small samples, unbiased-looking estimators may still be biased because of non-linear transformations (e.g., SD from variance).
- When multiple estimators exist for the same parameter, compare them by MSE, not just bias or variance alone.
- The Bayesian posterior mean is an estimator too – it optimises MSE among squared-error-loss rules under the posterior.
- Never report a point estimate as if it were a true parameter; the estimate is random, the parameter is not, and the inference must reflect that.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr